

NETFLIX RECOMMENDATION SYSTEM

PROBLEM STATEMENT & OBJECTIVE

GOAL

BUILD A PERSONALIZED RECOMMENDATION SYSTEM USING THE NETFLIX PRIZE DATASET CAPABLE OF:

- Learning user preferences from historical ratings
- Predicting unseen movie ratings
- Generating Top-K personalized recommendations
- Evaluating recommendation quality using RMSE and MAP@10

Dataset Challenges

- 100M+ RATINGS
- 480K+ USERS
- 17K+ MOVIES
- EXTREMELY SPARSE USER-ITEM MATRIX
- COLD-START USERS AND ITEMS
- TEMPORAL EVOLUTION OF PREFERENCES

Project Approach Compare:

- Item-Based Collaborative Filtering
- Matrix Factorization (SVD)
- SVD++

under multiple evaluation strategies.

Problem
Statement

Data
understanding

Evaluation
Strategy

Recommendations
models

Experimental
Results

Recommendation

Conclusion

DATA UNDERSTANDING & EDA

DATASET USED

ORIGINAL NETFLIX PRIZE
DATASET

Metric	Value
Ratings	100,480,507
Users	480,189
Movies	17,770

MODELING DATASET

DUE TO COMPUTATIONAL
CONSTRAINTS:

- SAMPLED 2,000,000 RATINGS
- 358,375 USERS
- 17,324 MOVIES
- 1999–2005 TIME PERIOD

Key EDA Findings

Rating Distribution

- Mean Rating = 3.61
- Median Rating = 4
- Majority ratings between 3 and 5

User Activity

- Most users have very few ratings
- Few highly active users dominate interactions

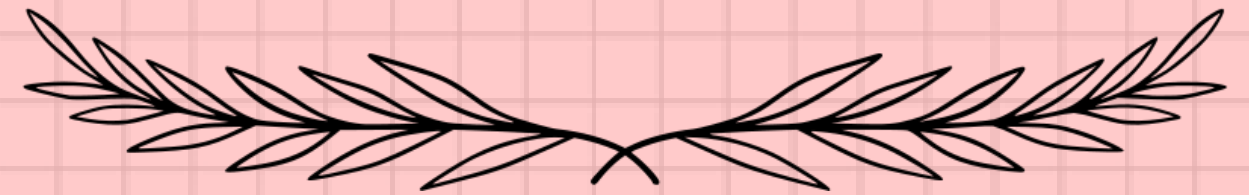
Movie Popularity

- Strong long-tail distribution
- Small set of movies receives most ratings

IMPORTANT INSIGHT
ALTHOUGH ONLY 2M RATINGS WERE
USED, THE SAMPLE RETAINED:

17,324 MOVIES OUT OF 17,770
≈ 97.5% MOVIE COVERAGE

MAKING IT HIGHLY REPRESENTATIVE
OF THE ORIGINAL CATALOG.



EVALUATION STRATEGY & SPLIT ANALYSIS

Why Multiple Train-Test Splits?

RANDOM SPLITTING OFTEN LEAKS FUTURE INFORMATION.
THEREFORE, THREE EVALUATION STRATEGIES WERE
EXPLORED.

1. RANDOM SPLIT

80% TRAIN
20% TEST

PROS:

- EASY BENCHMARK

CONS:

- UNREALISTIC FUTURE
PREDICTION

2. GLOBAL TEMPORAL SPLIT

PAST RATINGS → TRAIN
FUTURE RATINGS → TEST

MORE REALISTIC FOR
PRODUCTION SYSTEMS.

3. PER-USER TEMPORAL SPLIT

FOR EVERY USER:

OLD RATINGS → TRAIN
RECENT RATINGS → TEST

MOST REALISTIC
EVALUATION.

Important Observation

Cold-start analysis revealed:

Unseen Test Users = 57,336

Cold User Ratings $\approx$ 48%

showing why recommendation is difficult in real-world settings.

Problem Statement	Data understanding	Evaluation Strategy	Recommendations models	Experimental Results	Recommendation	Conclusion
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RECOMMENDATION MODELS						
MODEL 1: ITEM-BASED COLLABORATIVE FILTERING				MODEL 2: SINGULAR VALUE DECOMPOSITION (SVD)		
IDEA				IDEA		
RECOMMEND MOVIES SIMILAR TO THOSE ALREADY LIKED BY A USER.				LEARN HIDDEN LATENT FACTORS REPRESENTING: USER PREFERENCES MOVIE CHARACTERISTICS		
METHOD				BENEFITS		
USER-ITEM RATING MATRIX COSINE SIMILARITY NEIGHBORHOOD-BASED PREDICTION				HANDLES SPARSITY BETTER CAPTURES HIDDEN RELATIONSHIPS		
ADVANTAGES						
SIMPLE EXPLAINABLE				MODEL 3: SVD++		
LIMITATIONS				EXTENSION OF SVD USING IMPLICIT FEEDBACK FROM USER INTERACTIONS.		
SPARSE DATA PROBLEM POOR SCALABILITY						
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EXPERIMENTAL RESULTS (RMSE)

Rating Prediction Performance

- ModelRMSE
- Item-CF Random - 1.088
- SVD Random - 0.985
- Item-CF Temporal - 1.125
- SVD Temporal - 1.034
- Item-CF Per User Temporal - 1.154
- SVD Per User Temporal - 0.973
- Tuned SVD- 970
- SVD++ - 0.979

Hyperparameters of Best Model

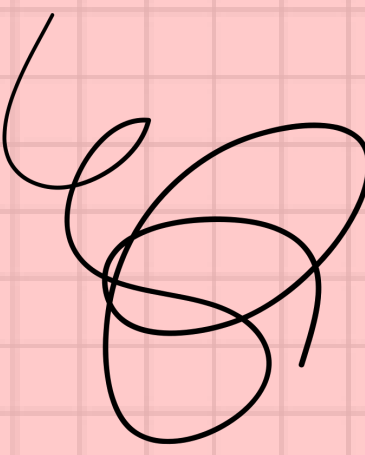
n_factors = 100
n_epochs = 50
learning_rate = 0.003
regularization = 0.05

Key Findings

- SVD CONSISTENTLY OUTPERFORMED ITEM-CF.
- TEMPORAL EVALUATION IS SIGNIFICANTLY HARDER THAN RANDOM EVALUATION.
- TUNED SVD ACHIEVED THE BEST PREDICTION ACCURACY.

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Recommendation Quality & Top-K Recommendations



Top-K Recommendation Generation
Recommendations generated using the best performing model (Tuned SVD).

Example Recommendations

BORCELLE

Movie	Predicted Rating
The West Wing: Season 3	4.75
Lord of the Rings: Return of the King	4.70
Spirited Away	4.69
The Sopranos: Season 3	4.67
Lost: Season 1	4.64

Observation

The model successfully recommends:

- Critically acclaimed movies
- Popular TV series
- Highly rated content indicating meaningful preference learning.

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Ranking Evaluation & Cold Start Analysis

Ranking Metrics
Original Evaluation
MetricScore



Precision@10 - 0.594
Recall@10 - 0.999
MAP@10 - 0.644

Final Evaluation
MetricScore



Precision@10 - 0.437
Recall@10 - 0.898
MAP@10 - 0.597

Conclusions

Best Model
Tuned SVD
RMSE = 0.970

Major Findings

- Matrix factorization outperformed collaborative filtering.
- Temporal evaluation provided more realistic performance estimates.
- Recommendation quality remained strong under stricter evaluation.
- The sampled dataset preserved almost the entire movie catalog.
- Cold-start users remain challenging.

Cold Start Results
SegmentRMSE



Cold - 1.697
Light- 1.737
Medium - 1.910
Heavy - 3.211

Issue Identified

Many users had very few test interactions.
To obtain a more realistic estimate:
Only users with ≥ 10 test ratings were evaluated.

Future improvements

- Alternating Least Squares (ALS)
- Neural Collaborative Filtering
- Hybrid Recommendation Systems
- Content-Based Features
- Full Netflix Dataset Training

Final Takeaway

The project demonstrates that latent-factor models such as SVD can effectively capture user preferences and generate meaningful recommendations at scale while balancing predictive accuracy and ranking quality.

Insight

Cold-start remains one of the most significant challenges for recommendation systems.

Problem Statement	Data understanding	Evaluation Strategy	Recommendations models	Experimental Results	Recommendation	Conclusion➤
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Thank you

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